INCH-POUND

MIL-C-46694A (AR)
AMENDMENT 6
7 July 1992
SUPERSEDING
AMENDMENT 5
02 March 1990

MILITARY SPECIFICATION

CHARGE, PROPELLING, 8 INCH, M1 (GREEN BAG)
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-C-46694A (MU) dated 28 April 1964, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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2.1 Government documents. Delete DRAWING "76-5-85" and substitute "8880530 - Container, Metal, Propelling Charge: M18A2". Under PUBLICATIONS delete "IEL-6119358 - Index of Inspection Equipment Lists".

3.4 Delete "Propelling charge increment weight. The charge increments weights" and substitute "Increment loading assembly weight. The net weight of the propellant."

3.4.1 Charge, propelling assembly weight. Delete in its entirety and substitute the following:

"3.4.1 Charge, propelling assembly weight. The charge, propelling assembly weight shall not differ by more than ± .50 ounces from the total of the amount specified in the loading authorization plus, average or actual weights of the empty increment bag and loaded igniter assembly (see 6.5)."

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Add the following paragraphs:

"3.7 Proving Ground

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3.7.1 Muzzle velocity. At 70 degrees Fahrenheit (F) the muzzle velocity shall be 1350 feet per second (fps) minimum, and 1410 fps maximum and the velocity standard deviation shall not exceed 5 fps.

3.7.2 Pressure. The maximum individual pressure at 70 degrees F shall not exceed 35,000 pounds per square inch.

3.8 Test plug assembly torque. The test plug shall be assembled to withstand subsequent application of the minimum assembly torque specified on the applicable drawings, without moving. (see 4.2.2.5).

3.9 Initial production lot inspection. This specification makes provisions for initial production lot inspection. Requirements for the submission of production samples by the contractor shall be as specified in the contract. (see 6.6).

* 3.10 First article test. Test samples shall be withdrawn from the first lot produced by the contractor and tested in accordance with all the requirements of this specification and with Table I (see 6.7). Until the first article test has been performed and found to be acceptable, the contractor is in know way authorized to initiate production of a second lot unless otherwise directed by the contracting officer."

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Add the following paragraphs:

"4.1.4 Initial production lot inspection.

4.1.4.1 Submission. The contractor shall submit an initial production lot sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.1.4.2. The initial lot sample shall consist of twenty-four (24) propelling charges packed in metal containers which have been produced by the contractor and selected randomly from the first week's regular production. A sample containing known defects will not be submitted.

4.1.4.2 Inspections to be performed. Propelling charges will be subjected by the Government to any or all the examinations or tests specified in 4.2.2 and 4.2.3 of this specification and any or all requirements of the applicable drawings.

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4.1.4.3. Rejection. If any assembly fails to comply with any of the applicable requirements, the initial production lot sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly to comply with any of the stated requirements."

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* 4.2.1a, Add the following to the end of the paragraph:

"An entire lot of propellant shall be utilized in the formation of one lot of propelling charges unless otherwise approved by the contracting officer."

4.2.2.1:

Critical category 1: After "missing" add "or damaged to the extent that black powder can escape without force."

Major category 101: Delete in its entirety and substitute "101. Assembly damaged to the extent that propellant can escape without force.....Visual 01002."

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4.2.2.2: Major category 101: After "escape" add "without force".

4.2.2.3:

Critical category: Delete "None defined" and substitute, "1. Assembly damaged to the extent that black powder can escape without force.....Visual 03008."

Major category: Delete "AQL 0.65" and substitute "AQL 1.00"

Major category: 103: Delete in its entirety and substitute the following:

"103. Assembly damaged to the extent that propellant can escape without force.....Visual 03003"

Add the following:

"105, Increment missing or improperly assembled.....
....Visual 03009"

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4.2.2.4:

DELETE: "dwg 76-5-85" and substitute "8880530".

Delete: Major Defect 102 in its entirety.

Delete: Minor Defect 201 in its entirety.

Add Major Defect 103:

"103. Charge cannot be removed by hand (Note A) 4.4 04003"

Add Note:

"Note A - One sample from every eight hours production."

4.2.2.5:

Delete dwg. "76-5-85" and substitute "8880530".

Major category: Delete "none defined" and substitute the following:

"Major 100 percent
101. Test plug missing.....Visual 05006

Major AQL 0.40 percent
102. Torque test failure of plug (see 3.8).....Test 05007"

Minor category 201: Delete "Missing or".

Minor category 202: Delete in its entirety.

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4.2.3.2: Propelling Charge (increment weight). Delete in its entirety and substitute the following:

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"4.2.3.2 Increment Loading Assembly Weight (see 3.4). Major defect. - The increment shall be weighed and check-weighed 100 percent. Any increment which fails to comply with the applicable requirements shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.2.4.

4.2.3.2.1 Method 1. The propellant shall be weighed in a container and placed in the bag, the bag sewn and the increment loading assembly weighed. The net weight of the propellant may be calculated using an average tare weight of the empty bags or the tare weight as marked on the individual bag.

4.2.3.2.2 Method 2. The propellant in a container shall be weighed and then check-weighed on a different balance and if performed manually, by another operator. The propellant shall then be placed in the bag and the bag sewn. The weight of the loaded bags should be checked as a major defect using a sampling plan from MIL-STD-105 or MIL-STD-1235 with an AQL of 0.40 percent. The net weight of the propellant shall be calculated using an average weight of the empty bags or the tare weight as marked on the individual bag."

Add the following paragraphs:

"4.2.3.4 Charge, Propelling Assembly Weight. (See 3.4.1) Major Defect. - The assembly shall be weighed 100 percent. Any assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

4.2.3.5 Proving Ground (See 3.7).

4.2.3.5.1 Muzzle Velocity: Major defect: Code No. 08001. The sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B3, Code Letter D, and an AQL of 0.65 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.65 exceeds the applicable requirement, the lot shall be rejected.

4.2.3.5.2 Retest. If for any reason the Proving Ground considers that the test conditions have detrimentally affected the test results, they shall declare the results invalid and perform a new test on addition sample charges as required."

Add the following paragraph:

"4.2.3.6 Weight of Black Powder (see dwg 8860504) Major defect - The weight of black powder in the igniter assembly shall be inspected 100 percent as specified in 4.3.4."

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4.2.4: Inspection equipment. Delete in its entirety and substitute the following:

"4.2.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed in this section is identified under the "Methods of Inspection" headings in 4.2 and 4.3 (see 6.3)."

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4.2.4.2: Supply and maintenance. Delete in its entirety.

4.3.2 Air pressure test for container. Delete in its entirety and substitute the following:

"4.3.2 Air pressure test. Each container shall be tested as specified on the appropriate drawing. A pressure drop equivalent to or greater than the applicable requirement shall be cause for rejection and removal from the lot."

Add the following paragraphs:

"4.3.3 Proving Ground. This test shall be conducted at a Government Proving Ground in accordance with the applicable Acceptance Test Procedure. The sample charges shall be fired in an M2 howitzer for which the charge is standard and observation shall be made for conformance to all applicable requirements. The charges shall be conditioned at 70 degrees Fahrenheit ± 2.5 degrees Fahrenheit for 24 hours and fired at Zone 5 using inert loaded M106 projectiles of the standard weight zone and dummy fuzes. Calibration rounds will be used for muzzle velocity and chamber pressure correction.

4.3.4 Weight of Black Powder in Base Igniter Assembly. The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. The weight of the black powder in the igniter will be determined by subtracting an average weight of the empty assembly or the actual weight as marked on the individual assembly from the loaded igniter (no propellant) and base loading assembly weight. Any assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

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4.4 Charge removal from container. Each sample shall be tested for removal of the propelling charge from the container by hand. The charge may be pulled out by the straps or the container may be inverted and the charge either pulled out or slid out. Any charge that cannot be removed from the container using the above method shall be cause for rejection of the production represented by the sample."

* 5. Delete 5, 5.1, 5.1.1, 5.2, 5.2.1, and 5.3 in their entirety and substitute the following:

"5 PACKAGING.

5.1 Preservation and packaging. N/A.

5.2 Packing.

5.2.1 Level A. The propelling charge shall be packed in a container complying with dwg. 8880530. Containers shall be closed, sealed and tested in accordance with the requirements of dwg. 7548187.

5.2.2 Level B. Pack the same as Level A.

5.2.3 Level C. Pack the same as Level A.

5.3 Marking. Marking shall be in accordance with specified requirements of drawing 7548187."

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6.3, 6.3.1, 6.3.2, 6.3.3, 6.3.4, 6.3.5, 6.3.6, and 6.3.7: Delete in their entirety and substitute the following:

"6.3 Inspection Equipment Designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) design and contractor designs. SIE designs are designated by drawing numbers or under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs. SIE designs may consist of any of the following:

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a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any specified fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and force Requirements List (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Designs" for guidance. The specification number and other specific identifying information from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal review will normally be accomplished within one month after receipt.

6.5 Charge, Propelling Assembly Weight. This requirement applies to the load plant only due to change in tare weight of

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the bags because of variations in bag material and moisture content.

6.6 Initial Production Lot Procedures. Sample propelling charges shall be submitted to Commander, Picatinny Arsenal, New Jersey 07806-5000, ATTN: SMCAR-QAR-R upon implementation of this Military Specification for those items procured for the U.S. Army."

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* Add the following paragraph:

"6.7 First article test. The first article test is a one time qualification test for a new contractor to load assemble and pack acceptable M1 propelling charges. Once a contractor has demonstrated the ability to load, assemble and pack the propelling charges, the first article test may be waived by the contracting officer. The following test program shall be used for the first article test:

Table I
First Article Test

Test Phase	Temp (deg F)	No. Control Rounds	No. Test Rounds	TECOM TOP
	70		10	TOP-4-2-504
Propellant	145		10	TOP-4-2-504
checkout	-60		10	TOP-4-2-504
Sequential	145	15	60	TOP-4-2-504
Environmental	-60	15	60	TOP-4-2-504"

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